



**Universität
Zürich** ^{UZH}

Feb 27, 2023

Ihsan Ul Haq

has successfully completed

An Intuitive Introduction to Probability

an online non-credit course authorized by University of Zurich and offered through
Coursera

A handwritten signature in black ink, reading 'Karl Schmedders'.

Karl Schmedders
Professor
Business Administration

**COURSE
CERTIFICATE**



Verify at:
<https://coursera.org/verify/8BCBVT8NRJJX>

Coursera has confirmed the identity of this individual and
their participation in the course.



May 25, 2023

Ihsan Ul Haq

has successfully completed

Convolutional Neural Networks

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

A handwritten signature in black ink, appearing to read 'Andrew Ng'.

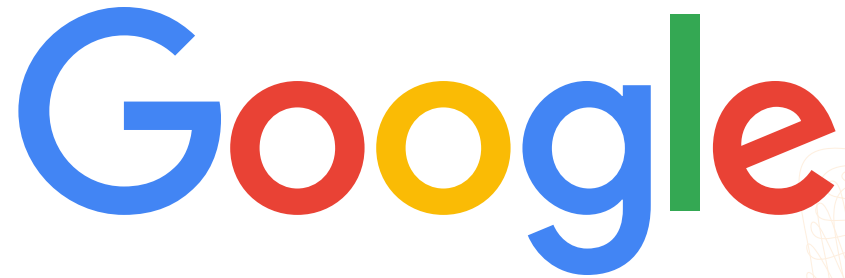
Andrew Ng, Founder, DeepLearning.AI & Co-founder, Coursera

COURSE
CERTIFICATE



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Coursera has confirmed the identity of this individual and
their participation in the course.



Feb 2, 2023

Ihsan Ul Haq

has successfully completed

Crash Course on Python

an online non-credit course authorized by Google and offered through Coursera

Google

Google

COURSE
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Coursera has confirmed the identity of this individual and their participation in the course.



UNIVERSITY OF ALBERTA



Feb 8, 2023

Ihsan Ul Haq

has successfully completed

Fundamentals of Reinforcement Learning

an online non-credit course authorized by University of Alberta, Alberta Machine Intelligence Institute and offered through Coursera

Martha White
Assistant Professor
Computing Science
Faculty of Science

Adam White
Assistant Professor
Computing Science
Faculty of Science

COURSE CERTIFICATE



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<https://coursera.org/verify/3NC8GM7XMD6Q>

Coursera has confirmed the identity of this individual and their participation in the course.



Jan 7, 2026

Ihsan Ul Haq

has successfully completed

Generative AI with Large Language Models

an online non-credit course authorized by DeepLearning.AI and Amazon Web Services
and offered through Coursera

Handwritten signatures of Antje Barth, Chris Fregly, and Mike Chambers in black ink.

Antje Barth, Chris Fregly
Shelbee Eigenbrode, Mike Chambers

COURSE
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<https://coursera.org/verify/RFNU70UAG6XI>

Coursera has confirmed the identity of this individual and
their participation in the course.



May 21, 2023

Ihsan Ul Haq

has successfully completed

Improving Deep Neural Networks: Hyperparameter
Tuning, Regularization and Optimization

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

A handwritten signature in black ink, appearing to read 'Andrew Ng'.

Andrew Ng, Founder, DeepLearning.AI & Co-founder, Coursera

COURSE
CERTIFICATE



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Coursera has confirmed the identity of this individual and
their participation in the course.



Feb 22, 2023

Ihsan Ul Haq

has successfully completed

Introduction to TensorFlow for Artificial
Intelligence, Machine Learning, and Deep Learning

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

A stylized, handwritten signature in blue ink that reads 'Laurence Moroney'.

Laurence Moroney
Instructor

COURSE
CERTIFICATE



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Coursera has confirmed the identity of this individual and
their participation in the course.



Feb 17, 2023

Ihsan Ul Haq

has successfully completed

Linear Algebra for Machine Learning and Data Science

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

A handwritten signature in black ink that reads 'Luis Serrano'.

Luis Serrano, Instructor, Serrano Academy. Anshuman Singh, Curriculum Product Manager, DeepLearning.AI. Elena Sanina, Curriculum Engineer, DeepLearning.AI. Magdalena Bouza, Curriculum Developer, DeepLearning.AI. Obed Nsiah, Curriculum Developer. Lucas Coutinho, Curriculum Developer, DeepLearning.AI

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IMPERIAL

May 22, 2026

Ihsan Ul Haq

has successfully completed

Mathematics for Machine Learning: Linear Algebra

an online course authorized by Imperial College London and offered through Coursera



David Dye, Samuel J. Cooper, and A. Freddie Page

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This certificate attests to the learner's completion of an online course / project delivered via Coursera. It does not constitute formal enrollment at any university or entity and does not itself grant academic credit, grades, or a degree. Institutions or organizations may, at their discretion, recognize this learning toward their own programs or credentials.

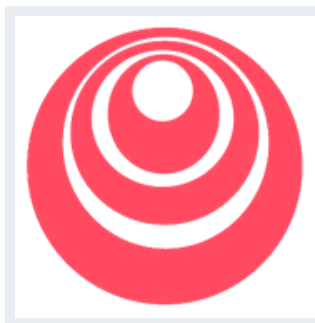


3 Courses

Supervised Machine Learning: Regression and Classification

Advanced Learning Algorithms

Unsupervised Learning, Recommenders, Reinforcement Learning



Dec 19, 2022

Ihsan Ul Haq

has successfully completed the online, non-credit Specialization

Machine Learning

Congratulations on completing all three courses of the Machine Learning Specialization! You studied modern machine learning concepts, including supervised learning (linear regression, logistic regression, neural networks, decision trees), unsupervised learning (clustering, anomaly detection), recommender systems, and reinforcement learning. You learned some of the best practices for building machine learning models. You've also gained practical skills to apply machine learning techniques to challenging real-world problems. Now #BreakIntoAI and start building your career in machine learning!

Andrew Ng, Instructor,
DeepLearning.AI
Eddy Shyu, Curriculum
Architect,
DeepLearning.AI
Aarti Bagul and Geoff
Ladwig, Curriculum
Engineers,
DeepLearning.AI

The online specialization named in this certificate may draw on material from courses taught on-campus, but the included courses are not equivalent to on-campus courses. Participation in this online specialization does not constitute enrollment at this university. This certificate does not confer a University grade, course credit or degree, and it does not verify the identity of the learner.

Verify this certificate at:

<https://coursera.org/verify/specialization/4MKZ5RLEJSKV>



May 19, 2023

Ihsan Ul Haq

has successfully completed

Neural Networks and Deep Learning

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

A handwritten signature in black ink, appearing to read 'Andrew Ng'.

Andrew Ng, Founder, DeepLearning.AI & Co-founder, Coursera

COURSE
CERTIFICATE



Verify at:
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Coursera has confirmed the identity of this individual and
their participation in the course.



Feb 6, 2023

Ihsan Ul Haq

has successfully completed

Python Data Structures

an online non-credit course authorized by University of Michigan and offered through
Coursera

A handwritten signature in black ink, appearing to read 'Charles', followed by a horizontal line.

Charles Severance
Clinical Professor, School of Information
University of Michigan

COURSE
CERTIFICATE



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Coursera has confirmed the identity of this individual and
their participation in the course.



Jun 11, 2023

Ihsan Ul Haq

has successfully completed

Python for Data Science, AI & Development

an online non-credit course authorized by IBM and offered through Coursera

A handwritten signature in black ink, reading "Joseph Santarcangelo".

Joseph Santarcangelo
Senior Data Scientist
IBM

COURSE
CERTIFICATE



Verify at:
<https://coursera.org/verify/D6FWB9Y69XWE>

Coursera has confirmed the identity of this individual and
their participation in the course.



University of Colorado
Boulder

Dec 3, 2023

Ihsan Ul Haq

has successfully completed

Renewable Energy Technology Fundamentals

an online non-credit course authorized by University of Colorado Boulder and offered
through Coursera

A handwritten signature in black ink, reading "Stephen R. Lawrence".

Stephen R. Lawrence, PhD
Associate Professor Emeritus
Leeds School of Business

COURSE
CERTIFICATE



Verify at:
<https://coursera.org/verify/BEPL8YRY4NVH>

Coursera has confirmed the identity of this individual and
their participation in the course.



University of Colorado
Boulder

Jan 12, 2024

Ihsan Ul Haq

has successfully completed

Renewable Power and Electricity Systems

an online non-credit course authorized by University of Colorado Boulder and offered
through Coursera

A handwritten signature in black ink that reads "Paul Komor".

Paul Komor
Research Faculty
Energy Institute
University of Colorado Boulder

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CERTIFICATE



Verify at:

<https://coursera.org/verify/3DZMGN5K276N>

Coursera has confirmed the identity of this individual and
their participation in the course.



May 22, 2023

Ihsan Ul Haq

has successfully completed

Structuring Machine Learning Projects

an online non-credit course authorized by DeepLearning.AI and offered through Coursera

A handwritten signature in black ink, appearing to read 'Andrew Ng'.

Andrew Ng, Founder, DeepLearning.AI & Co-founder, Coursera

COURSE
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Verify at:
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Coursera has confirmed the identity of this individual and
their participation in the course.



May 26, 2025

Ihsan Ul Haq

has successfully completed

What is Data Science?

an online non-credit course authorized by IBM and offered through Coursera

A handwritten signature in black ink, appearing to read "Rav Ahuja".

Rav Ahuja
Global Program Director,
Skills Network

A handwritten signature in black ink, appearing to read "Alex Akison".

Alex Akison, Ph.D.
Data Scientist

COURSE
CERTIFICATE



Verify at:
<https://coursera.org/verify/L0F796I3LK48>

Coursera has confirmed the identity of this individual and
their participation in the course.